

Project Timeline

Project: A Mechanism-Informed Gene Signature for Regenerative vs. Fibrotic Tendon Healing

Stages at a Glance

Stage	Task	Status
0	Define project scope & research questions	Done ✓
0	Literature review (Harvey, Cherief, Howell, Kaji, Moser)	Done ✓
1a	Download & QC Cherief 2023 scRNA-seq (GSE244921)	Done ✓
1b	Clustering & cell-type annotation (Cherief 2023)	Done ✓
1c	Sub-cluster cluster 8 → TSPC / T-FAP separation	Done ✓
2	Obtain Harvey 2019 count matrix (Cell Ranger on FASTQs)	Done ✓
3a	pySCENIC — TF network inference on TSPC / T-FAP	Done ✓
3b	Label transfer Harvey 2019 → Cherief 2023 cluster 8	On hold
4	CellRank — fate trajectory from PDGFR α ⁺ progenitor	Done ✓ (pseudotime caveat)
5	LIANA — cell-cell communication (innervated vs. denervated)	Done ✓
6	Mechanistic summary (TF regulons + upstream signals)	Not started ← next
7	Feature selection (LASSO + Boruta + XGBoost)	Not started
8	Classifier training & validation	Not started
9	Continuous healing index score	Not started

Week 1 — May 04–10

Focus: Project orientation & literature

- Sourced and uploaded key reference papers (Harvey 2019, Cherief 2023, Howell 2017, Kaji 2020, Moser 2021) and seminar materials
- Began reading foundational literature on tendon fate mapping and PDGFR α + progenitor biology

Week 2 — May 11–17

Focus: Research framing

- Drafted initial project ideas and conceptual framework
- Outlined the two-arm structure: mechanistic arm (TF networks / CellRank) + applied arm (gene signature classifier)

Week 3 — May 18–24

Focus: Research plan & Cherief 2023 analysis

- Drafted proposal with detailed scientific rationale and pipeline stages
- Set up project directory structure
- Began working on Stage 1a: download and initial QC of Cherief 2023 scRNA-seq (GSE244921)
- BIN6999 course submissions: proposal, LLM agreement form, quality of research effort, setting expectations, transferable skills checklist

Week 4 — May 25–31

Focus: Cherief 2023 analysis

- Completed Stage 1a: QC filtering of GSE244921; retained 22,615 cells
- Completed Stage 1b: PCA → UMAP → Leiden clustering (resolution = 0.5); identified 17 clusters; annotated cell types by marker expression (tenocytes, macrophage subtypes, PDGFR α + stromal cluster 8)

- Completed Stage 1c: Sub-clustered cluster 8; swept Leiden resolutions 0.8–1.5; confirmed TSPC (Tppp3+) and T-FAP (Pdgfra+/Tppp3–) sub-populations; ran differential expression between sub-clusters
- Began working on Stage 2: Harvey 2019 data acquisition on HPC

Week 5 — Jun 01–07

Focus: Harvey 2019 analysis

- Major batch commit — completed Stage 2:
 - Downloaded SRR9087252 FASTQs (~23.2 GB) from SRA via bash SLURM jobs
 - Ran Cell Ranger 10 count on HPC; estimated 8,484 cells
 - QC-filtered to 4,069 cells (min 700 genes, <15% MT, <5,000 genes), matching Harvey 2019 paper
 - Normalized, selected 2,000 HVGs, PCA → UMAP → Leiden (res = 0.5); plotted fate markers (Tppp3, Pi16, Sfrp2, Pdgfra, Tnmd, Scx)

Week 6 — Jun 08–14

Focus: Harvey cluster annotation → pySCENIC setup

- Visited Scott Lab at Sunnybrook; got to see the wet lab
- Finished Harvey 2019 cluster annotation (notebook section 8): marker dotplot, cluster_annotation dict, re-saved .h5ad; **4,069 cells** retained across 12 clusters with TSPC and T-FAP annotated
- Set up pySCENIC HPC environment (Alliance Canada / NIBI); patched pySCENIC 0.12.0 for compatibility

Week 7 — Jun 15–21

Focus: pySCENIC TF network inference (Stage 3a)

- Ran full pySCENIC pipeline on HPC (GRNBoost2 → cisTarget → AUCell) on both Harvey 2019 and Cherief 2023 TSPC/T-FAP

- Produced **471 Harvey** and **417 Cherief** regulons; Wilcoxon + cross-dataset direction filter → **15 consistent T-FAP regulons** (AP-1, KLF, NF-κB families)
- **0 consistent TSPC regulons** — reflects homeostatic vs. post-injury context shift
- Stage 3a complete ✓; Stage 3b deferred (sub-cluster identities well-supported by markers; on hold as optional validation)

Week 8 — Jun 22–28

Focus: CellRank fate trajectory (Stage 4) + LIANA cell-cell communication (Stage 5)

- **CellRank (Stage 4):** PseudotimeKernel (DPT, no RNA velocity) on Cherief cluster 8 sub-clusters; GPCCA recovered all 4 macrostates; fate probability shift: TSPC 82.6% → 71.0% and T-FAP 17.2% → 28.9% (innervated → denervated); identified top lineage driver genes per fate; 119-gene pySCENIC × CellRank T-FAP overlap computed as mechanism-informed feature candidate set for Stage 7
- **CellRank caveat:** DPT pseudotime ordering unreliable (Tenogenic-progenitor ranks as most progenitor-like rather than Stromal); fate probabilities used as soft classification rather than mechanistic trajectory
- **LIANA (Stage 5):** rank_aggregate (6 methods, mouse LR database) on innervated vs. denervated; 11,669 LR pairs with TSPC/T-FAP as receiver; key findings — Sema3c/PDGF-C/Wnt5a/BMP3 lost at denervation; TNF-Notch + Sema3f + laminin-integrin gained; Efnal → EphA3 is the strongest differential signal overall
- Stages 4 and 5 complete ✓

Week 9 — Jun 29–Jul 05 (*current*)

Focus: Stage 6 — Mechanistic synthesis

- [] Integrate pySCENIC TF regulons, CellRank fate drivers, and LIANA upstream signals into a unified regulatory circuit
- [] Document signal-to-TF links (TNF → NF-κB; TGFβ → AP-1/ATF3; PDGF → MEK/ERK → AP-1) supported by GRNBoost2 adjacency tables
- [] Finalize the 119-gene T-FAP candidate feature set description and prepare for Stage 7

Upcoming

Week	Dates	Stage	Focus
Week 10	Jul 06–12	7–9	Feature selection + classifier training & validation + healing index score + figures finalized
Week 11	Jul 13–19	—	Buffer / storyboard polish; all analysis & storyboard due Jul 20
Week 12	Jul 20–26	—	Poster draft due Jul 27
Week 13	Jul 27–Aug 02	—	Poster revisions; final poster due Jul 31
Week 14	Aug 03–09	—	BIO2 Symposium — Aug 06
Week 15	Aug 10–16	—	Final report + self-reflection essay due Aug 11